DATA ANALYTICS PROJECT · 2020 DATA · REBUILT NOV 2026

Opioid Crisis Intelligence Dashboard

Does opioid prescribing explain overdose deaths across U.S. states, and which social factors matter more? A multi-source analysis of Medicare Part D, CDC and AHRQ data.

25.2M
CMS provider × drug rows processed

$R^2 = 0.006$
prescribing rate vs opioid death rate

$R^2 = 0.447$
with 5 social determinants added

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Programme	M.Sc. International Information Systems, FAU Erlangen-Nürnberg
Stack	PostgreSQL · Python (pandas, statsmodels, scipy, scikit-learn) · Power BI · GitHub Actions
Code and data	github.com/YajuvendraDhadhal/opioid-crisis-dashboard
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Executive summary

A state health department wants to know where to send opioid intervention resources. A common assumption is that states which prescribe more opioids also have more opioid deaths. This project tests that assumption with three public federal datasets for 2020 and builds a risk ranking of states.

56.9M
Medicare Part D opioid claims, 51 jurisdictions

54,781
opioid deaths in the 42 states with CDC data

West Virginia
highest combined risk score (82.8 of 100)

Key findings

- **Prescribing alone does not explain deaths.** Across 42 states, the Medicare opioid prescribing rate explains 0.6% of the variation in opioid death rates ($R^2 = 0.006$, $p = 0.61$).
- **Social factors explain much more.** Adding poverty, unemployment, uninsured rate, GINI index and median income raises R^2 to 0.447. Unemployment has a significant positive effect ($\beta = +0.41$, $p = 0.045$).
- **The uninsured rate is the strongest predictor, but with a negative sign** ($\beta = -0.50$, $p = 0.002$). States with many uninsured residents, such as Texas, Oklahoma and Georgia, had low opioid death rates in 2020. This most likely reflects geography, not a protective effect.
- **Top 10 priority states:** West Virginia, Kentucky, Tennessee, South Carolina, Mississippi, North Carolina, District of Columbia, Ohio, New Mexico, Missouri.
- **The drugs behind the crisis changed in 2020.** U.S. deaths involving synthetic opioids (mainly illicit fentanyl) rose 55% from 2019, and psychostimulants 48%, while heroin deaths fell 6%.

> **Recommendation in one line:** target help by economic distress and death rates, not by prescribing rates alone, and respond to illicit fentanyl and stimulants as well as to prescribing.

Business question

Which U.S. states show the strongest link between opioid prescribing and overdose mortality, and which socioeconomic factors amplify or buffer it, so that a state health department can prioritise intervention resources?

1 Data and method

Dataset	Source	Level	Used for
Medicare Part D Prescribers by Provider and Drug, 2020	CMS	Provider × drug, 25,209,729 rows	Opioid claims by state, drug and specialty
Medicare Part D Prescribers by Geography and Drug, 2020	CMS	State × drug	Official opioid drug flag
VSRR Provisional Drug Overdose Death Counts	CDC NCHS	State × month × drug indicator	Opioid deaths, 12 months ending Dec 2020; U.S. trend by drug type
Community-Level Health (SDOH) Database, 2020	AHRQ	County, 3,234 rows	Population, poverty, unemployment, uninsured, GINI, income

2020 is the most recent year in which all three sources overlap, so the design is a single-year comparison of states. Rates use the AHRQ county population, summed to state level.

Pipeline



The whole pipeline runs automatically on GitHub Actions with its own PostgreSQL container

Method

Step	What was done
Opioid definition	A drug counts as an opioid if most of its rows in the CMS Geography × Drug file have `Opioid_Drug_Flag = Y` (30 generic names)
State SDOH values	County values weighted by county population (SQL)
Rates	Opioid claims per 1,000 residents; opioid deaths per 100,000 residents
Correlation	Spearman, because the SDOH variables are right-skewed
Regression	OLS on standardized variables (comparable β), with VIF checks; gradient boosting as a non-linear cross-check
Risk score	30% prescribing + 40% death rate + 30% SDOH vulnerability, each scaled 0 to 100

2 Prescribing vs deaths

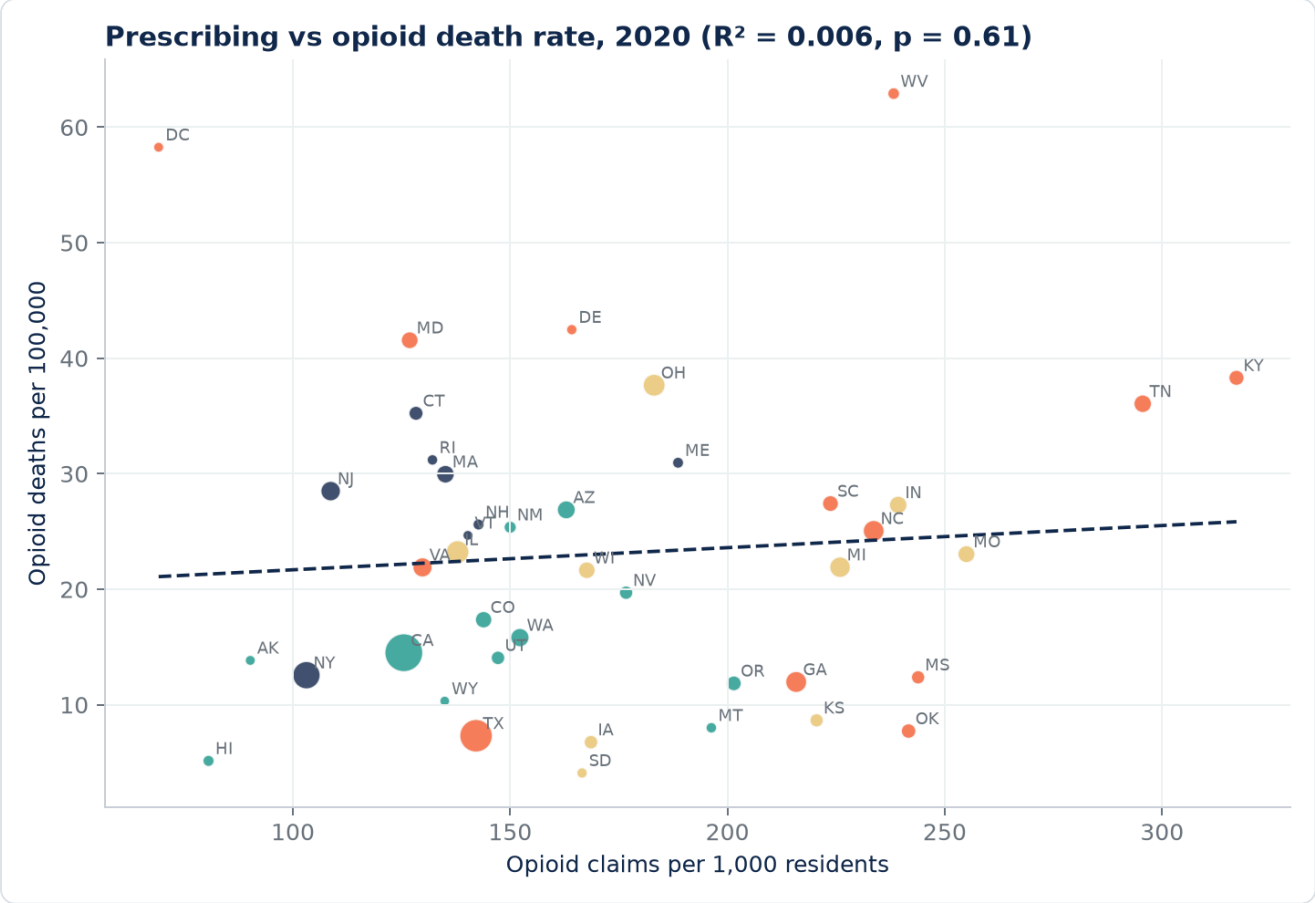


Figure 1. Each dot is a state (size = population). The trend line is almost flat.

States that prescribe more opioids to Medicare patients do not have higher opioid death rates. Kentucky and Tennessee prescribe the most and have high death rates. But West Virginia has a much higher death rate than Kentucky at lower prescribing, DC has one of the highest death rates at the lowest prescribing rate, and Mississippi and Oklahoma prescribe almost as much as West Virginia with far lower death rates.

3 What does explain deaths?

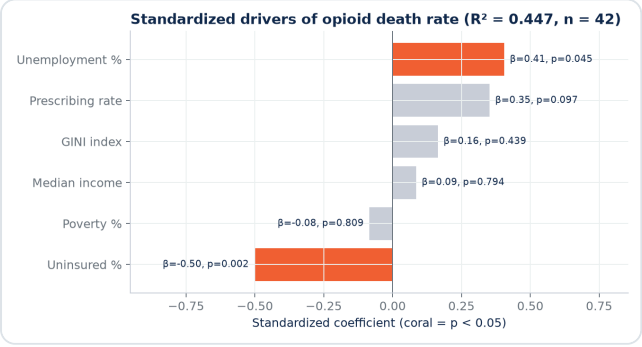
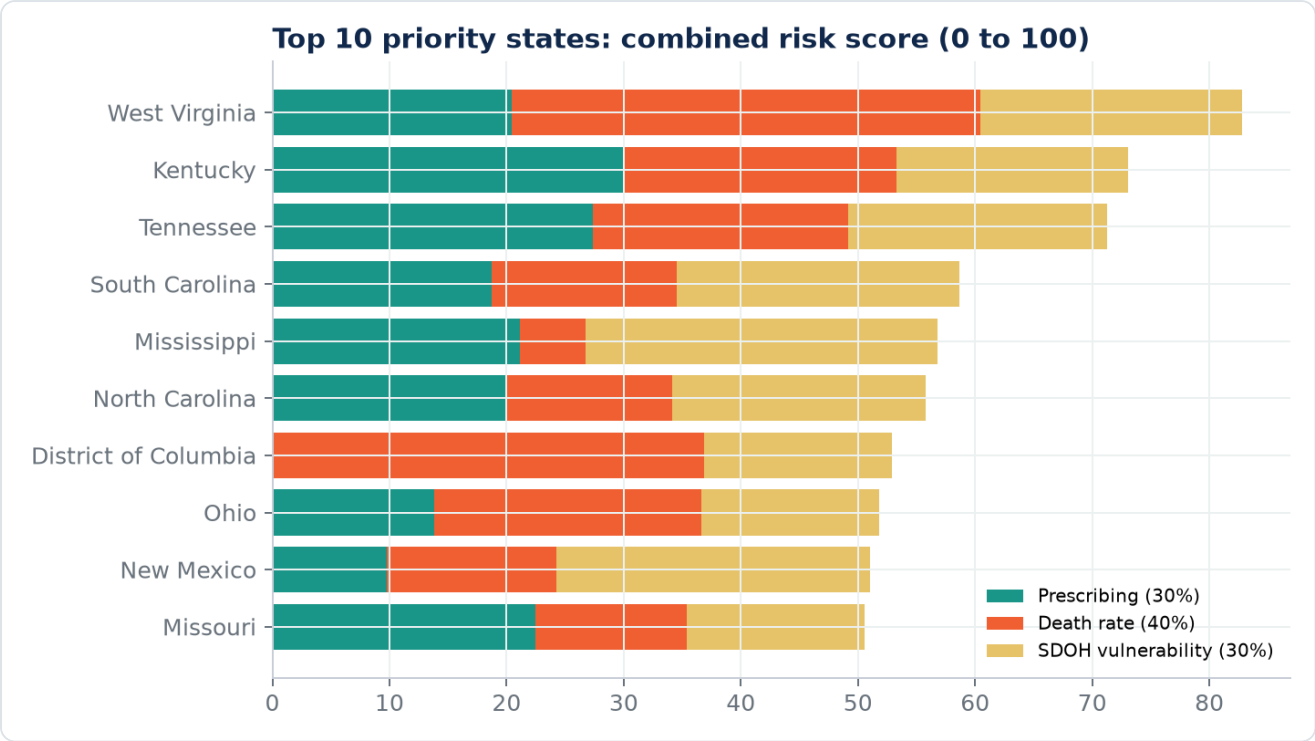


Figure 2. Standardized coefficients; coral = significant (p < 0.05).

Variable	β	p	VIF
Uninsured %	-0.50	0.002	1.4
Unemployment %	+0.41	0.045	2.4
Prescribing rate	+0.35	0.097	2.7
GINI index	+0.16	0.439	2.8
Median income	+0.09	0.794	6.7
Poverty %	-0.08	0.809	7.6

R^2 = 0.447, adjusted 0.352, F-test p = 0.001, n = 42. Poverty and income are strongly correlated (VIF 7.6 and 6.7), so their separate effects are not reliable.

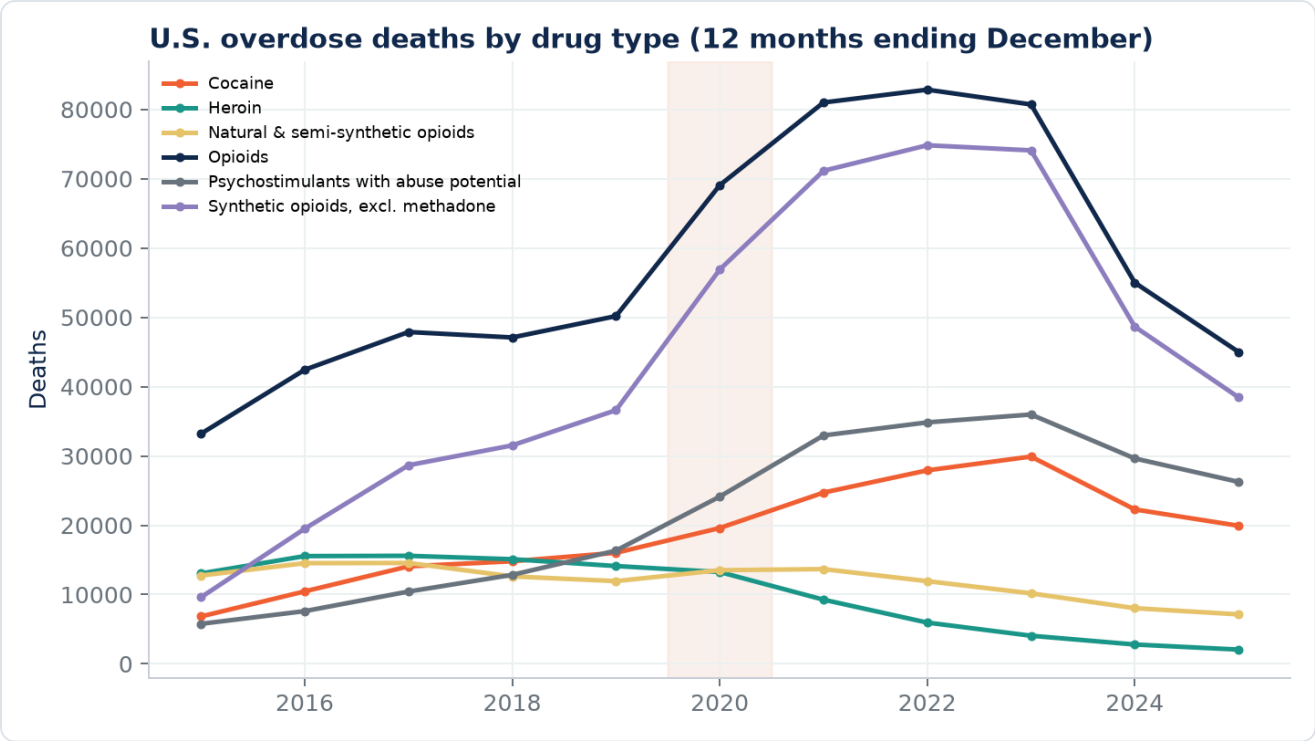
4 Top 10 priority states



Rank	State	Risk score	Deaths / 100k	Claims / 1k	SDOH score
1	West Virginia	82.8	62.9	238	74
2	Kentucky	73.1	38.3	317	66
3	Tennessee	71.3	36.1	296	74
4	South Carolina	58.7	27.4	224	80
5	Mississippi	56.8	12.4	244	100
6	North Carolina	55.8	25.1	234	72
7	District of Columbia	52.9	58.3	69	54
8	Ohio	51.8	37.7	183	51
9	New Mexico	51.0	25.4	150	89
10	Missouri	50.5	23.0	255	51

Seven of the ten are in the South. The District of Columbia ranks seventh with one of the lowest prescribing rates, because of its very high death rate (58.3 per 100,000), which shows why the death rate gets the largest weight.

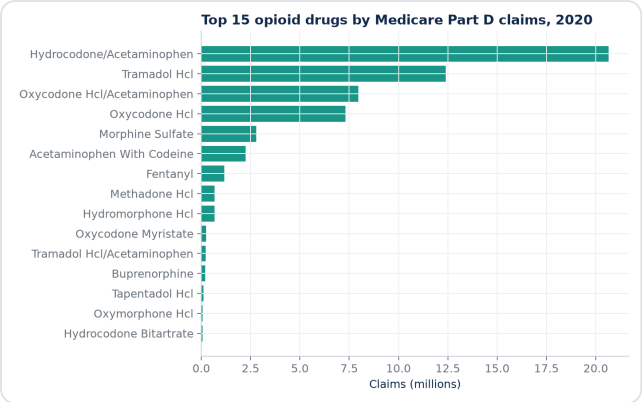
5 2020: the drugs behind the crisis

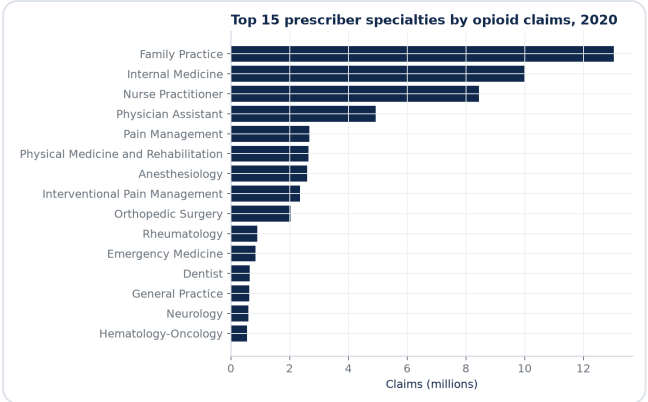


U.S. deaths, 12 months ending December	Change 2019 to 2020
Synthetic opioids excl. methadone (mainly fentanyl)	+55.4%
Psychostimulants (mainly methamphetamine)	+47.7%
All opioids	+37.6%
All drug overdose deaths	+30.0%
Cocaine	+22.3%
Natural and semi-synthetic opioids (prescription type)	+13.2%
Heroin	−6.1%

The fastest growth was in illicit fentanyl and stimulants, not in prescription-type opioids. This helps explain why the prescribing rate is a weak predictor of deaths by 2020.

6 Who prescribes, and what





Hydrocodone/acetaminophen (20.7M claims) and tramadol (12.4M) lead Medicare opioid prescribing. Family Practice (13.0M claims), Internal Medicine (10.0M) and Nurse Practitioners (8.4M) write the most opioid prescriptions. Pain Management has only 2,183 prescribers, but about 1,220 claims each, roughly six times the Family Practice average.

Social determinants across counties

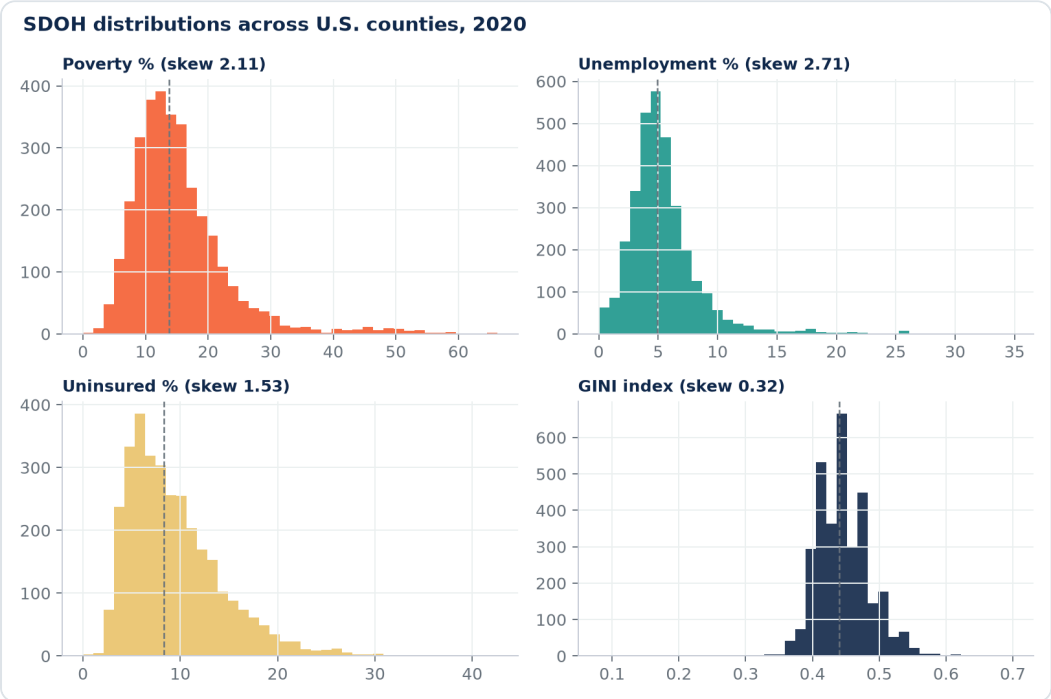
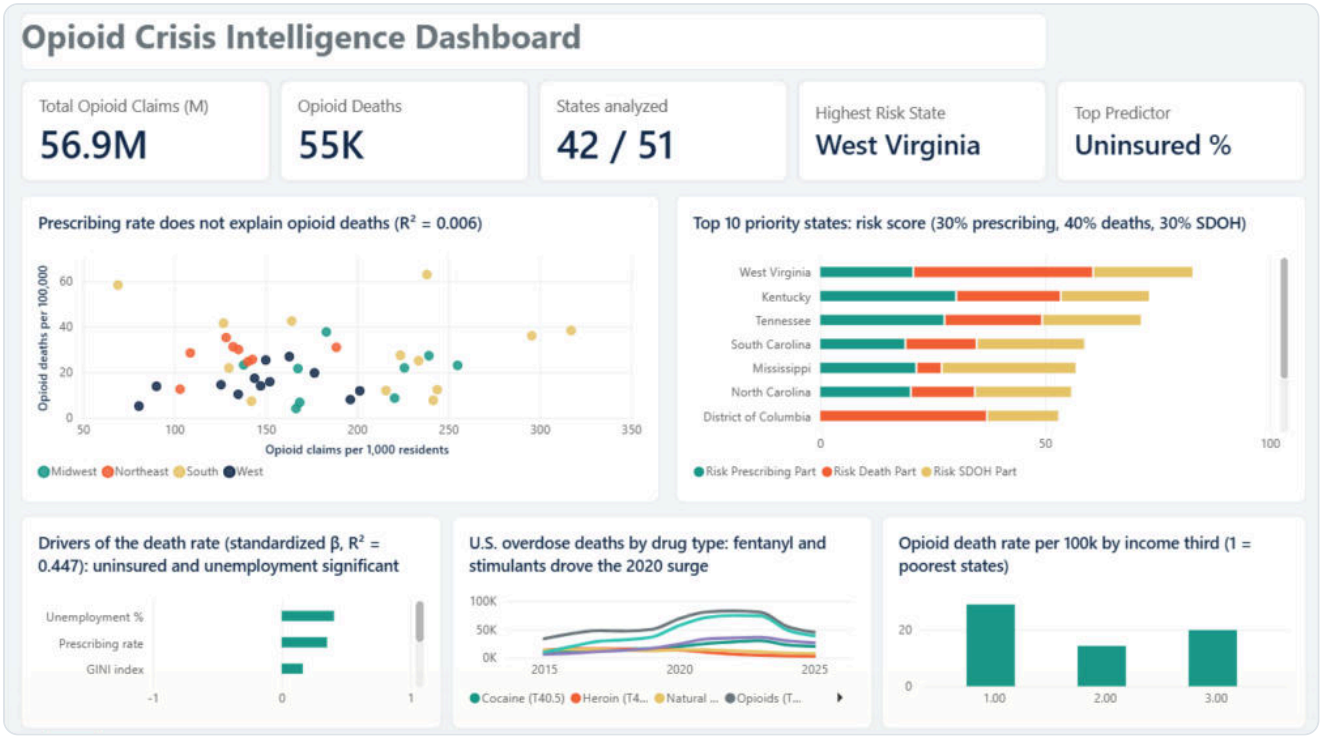


Figure. All four SDOH variables are right-skewed, with a long tail of disadvantaged counties. This is why Spearman correlation is used.

7 Power BI dashboard



One page, built in Power BI Desktop from the pipeline's star-schema tables (file: `powerbi/opioid_dashboard.pbix`).

Element	Content
5 KPI cards	Opioid claims, opioid deaths, states analysed, highest-risk state, strongest predictor (DAX measures)
Scatter plot	Prescribing rate vs death rate by state, coloured by region
Stacked bar	Top 10 states with the three parts of the risk score (Top N filter)
Bar chart	Standardized regression coefficients
Line chart	U.S. deaths by drug type, 2015 to 2025
Column chart	Death rate by income third of states

The semantic model (7 tables, 1 relationship, 17 measures) is stored as a TMDL script (`powerbi/model.tmdl`) so it can be recreated in any Power BI Desktop with one paste.

8 What went wrong

Problem	What I did
Inconsistent CMS opioid flag	Topiramate, an anti-seizure drug, appeared among the top opioids because 2 of its 11 rows are flagged as opioid. A drug now counts only if most of its rows are flagged. Buprenorphine HCl, mainly used to treat addiction, dropped out too. Claims went from 60.0M to 56.9M.
Numbers differ from my first version	My April 2026 version (code lost) reported R^2 = 0.043, 41 states, 58.8M claims and 11,680 deaths. This rebuild is fully reproducible from the raw files and finds R^2 = 0.006, 42 states, 56.9M claims and 54,781 deaths. The main conclusion is unchanged.
AHRQ download blocked	The server refused downloads without a browser user agent, and the file has moved to the new Community-Level Health Database. Fixed in `download.py`.
Map visuals disabled	Power BI's map visuals were switched off in the local security options, so the risk map slot shows the income comparison instead.

9 Limitations

- **Ecological data:** state-level patterns do not show individual causation.
- **Medicare covers mostly people over 65,** while most overdose deaths occur among 25 to 54 year olds.
- **9 states lack 2020 opioid death data** under the CDC indicator used (AL, AR, FL, ID, LA, MN, ND, NE, PA).
- **Small sample:** 42 states and 6 predictors; one year only, and 2020 was the first COVID-19 year.

10 Recommendations

1. **Prioritise distressed states with high death rates,** such as West Virginia, New Mexico and DC, instead of relying on prescribing rates.
2. **Keep prescriber education,** but target Family Practice, Internal Medicine and Nurse Practitioners, who write most opioid prescriptions.
3. **Respond to illicit fentanyl and stimulants:** naloxone, fentanyl test strips and drug checking.
4. **Close data gaps:** 9 states have no opioid-specific death data for 2020.